$$\gamma_k = E[R^2]\, E\left[\frac{1}{2}\left(\cos(2\pi f k) + \cos(2\pi(2t-k)+\phi)\right)\right]$$

$$= E[R^2]\left(\frac{1}{2}\cos(2\pi f k)\right) + \underbrace{E[\cos(2\pi(2t-k)+\phi)]}_{=0 \text{ cf (a)}}$$

the nul terms are justified by the fact that we showed in a) that $E[Y_t] = 0 \ \forall t$

finally :

$$\boxed{\gamma_k = \frac{E[R^2]}{2}\cos(2\pi f k)}$$